Finite p-Variation Jordan Curves and Inscribed Rectangles via Asano–Ike

Internal proof package — clean-room verified (Directive 007R)
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Abstract

We record an extension of the rectangular peg theorem to Jordan curves with finite p-variation for $1 \le p < 2$, combining Boedihardjo–Geng piecewise-linear interpolation [SOURCE], a velocity-matched cubic polygon-smoothing lemma T018 [LOCAL], Young primitive convergence [LOCAL], and Asano–Ike Theorem 1.1 [SOURCE]. Clean-room audit (Directive 007R) verified the proof chain. This document is **internally** verified only. We do **not** treat the critical $p = 2$ boundary (see §7).

1 Introduction

The Toeplitz square peg problem asks whether every Jordan curve in $\mathbb{R}^2$ inscribes a square. Asano and Ike (arXiv:2412.21057v3) prove that if a Jordan curve satisfies a continuous Legendrian-lift hypothesis (Theorem 1.1), then it inscribes every θ-rectangle for $\theta \in (0, \pi)$.

Theorem 1.1 (T017 [SYNTHESIS]). *Let $c : S^1 \to \mathbb{R}^2$ be a Jordan parametrization with finite p-variation for some $1 \le p < 2$. Then c satisfies the hypotheses of Asano–Ike Theorem 1.1 and therefore inscribes a θ-rectangle for every $\theta \in (0, \pi)$.*

Remark 1.2 (Novelty language). This appears to be an immediate synthesis of existing interpolation and microlocal peg results, with the controlled C^1 smoothing lemma supplied here. We do not claim priority without author confirmation.

2 Variation preliminaries [LOCAL]

Definition 2.1. For $u : [a, b] \to \mathbb{R}^d$,

$$|u|_{r\text{-var};[a,b]} = \Big(\sup_{\Pi} \sum_i |u(t_{i+1}) - u(t_i)|^r\Big)^{1/r}, \qquad \|u\|_\infty = \sup_t |u(t)|.$$

Constant offsets are not folded into the seminorm.

Lemma 2.2 (Interpolation [LOCAL]). *For $0 < p < q$ and finite p-variation h,*

$$|h|_{q\text{-}var} \le \operatorname{osc}(h)^{1-p/q} |h|_{p\text{-}var}^{p/q}, \quad \operatorname{osc}(h) = \sup_{s,t} |u(t) - u(s)|.$$

Lemma 2.3 (Variation monotonicity [LOCAL]). *For $q \ge 1$, $|h|_{q\text{-}var} \le |h|_{1\text{-}var}$.*

Proof. For each partition, $(\sum |\Delta h_i|^q)^{1/q} \le \sum |\Delta h_i|$ since $x^q \le x$ for $x \in [0, 1]$ applied after normalizing by $\max_i |\Delta h_i|$ when $\max_i |\Delta h_i| \le 1$; the general case uses $(\sum x_i^q)^{1/q} \le \sum x_i$ for nonnegative x_i. □

3 Polygonal interpolation [SOURCE]

Theorem 3.1 (Boedihardjo–Geng 2.2, $M = \mathbb{R}^2$ [SOURCE]). *For a Jordan curve γ and $\varepsilon > 0$, there is a partition $\mathcal{P}_\varepsilon$ with mesh $< \varepsilon$ such that the piecewise linear interpolant $\gamma^{\mathcal{P}_\varepsilon}$ is a simple closed polygon with vertices on γ in cyclic order on the same parameter domain.*

Lemma 3.2 (Boedihardjo–Geng 3.1, $f = \mathrm{id}$ [SOURCE]). *If γ has finite p-variation and $q > p$, then $\|\gamma - \gamma^{\mathcal{P}}\|_{q\text{-}var} \to 0$ as $|\mathcal{P}| \to 0$.*

4 T018: velocity-matched cubic smoothing [LOCAL]

Lemma 4.1 (T018 [LOCAL]). *Let $P : S^1 \to \mathbb{R}^2$ be a simple polygonal Jordan curve. For every $\varepsilon > 0$ and $q \geq 1$, there exists a regular C^1 Jordan map S with the same orientation and a common parametrization such that $\|S - P\|_\infty < \varepsilon$ and $\|S - P\|_{q\text{-}var} < \varepsilon$.*

Construction. At vertex $v = P(\tau)$, choose $\delta > 0$, $A = P(\tau - \delta)$, $C = P(\tau + \delta)$, $L = 2\delta$, $u_- = (v - A)/\delta$, $u_+ = (C - v)/\delta$. Cubic controls:

$$Q_0 = A,\ Q_1 = A + \tfrac{2}{3}(v - A),\ Q_2 = C - \tfrac{2}{3}(C - v),\ Q_3 = C.$$

On $[\tau - \delta, \tau + \delta]$, S is the cubic Bézier; elsewhere $S = P$.

Velocity matching. $\frac{dS}{dt}(\tau - \delta) = u_-$ and $\frac{dS}{dt}(\tau + \delta) = u_+$.

Injectivity. With $a = v - A$, $b = C - v$ non-antiparallel, $w = a/|a| + b/|b|$ satisfies $w \cdot dS/ds > 0$ on the corner arc.

Variation. $\mathrm{Var}_1(S - P; I_v) \leq 4r_v$ with $|a|, |b| \leq r_v$; sum over corners.

5 Young primitive convergence [LOCAL]

Write $c = (x, \xi)$, $\lambda = \xi\, dx$, $f(t) = \int_0^t \xi\, dx$ (Young, $q < 2$).

Theorem 5.1 (Young / BG 3.1 [SOURCE]). *If $1/p + 1/q > 1$, the Young integral exists; for $q < 2$, $2/q > 1$.*

With $\|S_n - c\|_{q\text{-var}} \to 0$,

$$\sup_{t \in [0, 2\pi]} |f_n(t) - f(t)| \leq C_q \|S_n - c\|_{q\text{-var}} \to 0.$$

Lift to $\mathbb{R}$ [LOCAL]. For $t = 2\pi k + s$, $|k| \leq N_K := \lceil K/(2\pi) \rceil + 1$,

$$\sup_{|t| \leq K} |f_n(t) - f(t)| \leq \eta_n + N_K |I_n - I| \to 0.$$

6 Application of Asano–Ike [SOURCE]

Normalize enclosed area to π. Theorem 1.1 (AI-PEG-v3) and Remark 5.6 (C^1 approximants suffice) yield all θ-rectangles. Depends on preprint arXiv:2412.21057v3.

7 Examples and limitations

The Koch snowflake is nonrectifiable with finite p-variation for $p > \log 4/\log 3$. We do not claim it lies outside the locally monotone class without a separate proof.

Unresolved $p = 2$ boundary [CONJECTURE]. At $q = 2$, Young requires $1/q+1/q = 1$ (not strict). Critical-modulus enhancement (Yang, arXiv:1204.0112) is treated separately in Directive 008; T020 remains conjectural.

References

[1] T. Asano and Y. Ike, *The rectifiable rectangular peg problem*, arXiv:2412.21057v3, 2026.

[2] H. Boedihardjo and X. Geng, arXiv:1309.1576v2.

[3] D. Yang, *Notes on area operator, geometric 2-rough paths and Young integral*, arXiv:1204.0112.